

Exponents and Powers

ICSE Mathematics — Class VII | Chapter Notes

This document is a cleanly reformatted, typed version of handwritten/scanned chapter pages, covering the theory, laws of exponents, worked examples, exercises (4.1–4.3), objective-type questions, MCQs, and a chapter summary with progress-check problems.

1. Introduction to Exponential Form

Look at the following: $3 \times 3 \times 3 \times 3 \times 3 = 3^5$, i.e., $3^5 = 243$.

Also, $2/3 \times 2/3 \times 2/3 \times 2/3 = (2/3)^4$, and $(-4) \times (-4) \times (-4) = (-4)^3$.

(2^5) is called the **exponential form** of 32. We have:

$32 = 2^5$ — here **2** is the **base** and **5** is the **exponent** (or **index**).

If a is any rational number and n is a natural number, then $a^n = a \times a \times a \times \dots$ (multiplied n times). This is read as 'a raised to the power n ' or 'a to the power n ' or simply 'a power n .' Here a is called the **base** and n is the **exponential form** of the number.

For example: $10^4 = 10 \times 10 \times 10 \times 10 = 10000$ [here base = 10, exponent (or index) = 4, and 10^4 is the exponential form of the number 10000].

Some power names

If the power of a number is 2, then it is named as its **square**, and if the power of a number is 3, then it is named as its **cube**. Thus:

$$9 = 3 \times 3 = 3^2 \text{ [read as 3 square or 3 raised to the power 2]}$$

$$100 = 10 \times 10 = 10^2 \text{ [read as 10 square or 10 raised to power 2]}$$

$$125 = 5 \times 5 \times 5 = 5^3 \text{ [read as 5 cube or 5 raised to power 3]}$$

$$1000 = 10 \times 10 \times 10 = 10^3 \text{ [read as 10 cube or 10 raised to power 3]}$$

Powers of -1

$(-1)^{\text{odd natural number}} = -1$ and $(-1)^{\text{even natural number}} = 1$. Thus, we have:

$$(-1)^5 = -1, (-1)^{100} = 1, \text{ and so on.}$$

2. Worked Examples — Base, Exponent & Values

Example 1. Write the base and exponent for the following exponential forms:

Solution. (i) For 5^3 , base = 5 and exponent = 3.

(ii) For 7^5 , base = 7 and exponent = 5.

(iii) For $(-3)^6$, base = -3 and exponent = 6.

Example 2. Find the value of the following:

Solution. (i) $a \times b$ is written as ab , read as 'a into b'.

(ii) $a \times a \times b \times b$ is written as a^2b^2 , read as 'a square into b square'.

Example 3. Simplify the following:

Solution. (i) $3 \times 3 \times 3 \times 3 = 3^4 = 81$

(ii) $2/3 \times 2/3 \times 2/3 \times 2/3 = (2/3)^4 = 16/81$

(iii) $(-4) \times (-4) \times (-4) = (-4)^3 = -64$

(iv) $(-4/5) \times (-4/5) \times (-4/5) \times (-4/5) = (-4/5)^4 = 256/625$

Exercise 4.1

1. Fill in the blanks:

- (i) In the expression 3^7 , base = ____, exponent = ____.
(ii) In the expression $(-7)^5$, base = ____ and exponent = ____.

2. Find the value of the expression $(2/5)^{11}$, base = ____ and exponent = ____.

3. Express the following in the exponential form:

- (i) $t \times t \times t$ (ii) $a \times a \times a \times a \times a$ (iii) $2 \times 2 \times x \times x \times x \times x \times x \times x \times x \times x \times c \times d$

4. Simplify the following:

- (i) $(2/5)^{-5}$ (ii) $(-3)^3 \times (-5)^2$ (iii) $25^2 \times (-1)^{31}$

5. Simplify the following:

- (i) $3^3 \times 9$ (ii) $(-3)^4 \times (-10)^4$ (iii) $4^2 \times 3^3 \times (-1)^{22}$

3. Laws of Exponents

We can multiply and divide rational numbers expressed in exponential form.

Multiplying powers with the same base

Let us find $2^4 \times 2^3$:

$$2^4 \times 2^3 = (2 \times 2 \times 2 \times 2) \times (2 \times 2 \times 2) = 2^{4+3} = 2^7$$

Law 1. If a is any rational number and m, n are natural numbers, then $a^m \times a^n = a^{m+n}$.

Dividing powers with the same base

Let us find $5^7 \div 5^4$:

$$5^7 \div 5^4 = 5^7/5^4 = (5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5)/(5 \times 5 \times 5 \times 5) = 5^{7-4} = 5^3$$

Law 2. If a is any non-zero rational number and m, n are natural numbers such that $m > n$, then $a^m \div a^n = a^{m-n}$.

Power of a power

Let us find $(2^3)^2$:

$$(2^3)^2 = 2^3 \times 2^3 = 2^{3+3} = 2^6 = 2^{3 \times 2}$$

Law 3. If a is any rational number and m, n are natural numbers, then $(a^m)^n = a^{m \times n}$.

Multiplying powers with the same exponent

Let us find $3^4 \times 5^4$:

$$3^4 \times 5^4 = (3 \times 3 \times 3 \times 3) \times (5 \times 5 \times 5 \times 5) = (3 \times 5) \times (3 \times 5) \times (3 \times 5) \times (3 \times 5) = (3 \times 5)^4$$

Law 4. If a, b are any rational numbers and n is a natural number, then $a^n \times b^n = (ab)^n$.

Dividing powers with the same exponent

Let us find $2^5/7^5$:

$$2^5/7^5 = (2 \times 2 \times 2 \times 2 \times 2)/(7 \times 7 \times 7 \times 7 \times 7) = (2/7)^5$$

Law 5. If a, b are any rational numbers ($b \neq 0$) and n is a natural number, then $a^n/b^n = (a/b)^n$.

Zero exponent

Let us find $3^4/3^4$. Since $3^4/3^4 = 1$, and also $3^4/3^4 = 3^{4-4} = 3^0$ (using Law 2), so both answers must be same.

Law 6. If a is any non-zero rational number, then $a^0 = 1$.

Negative exponent

Law 7. If a is any non-zero rational number, then $a^{-n} = 1/a^n$ (equivalently, $a^{-1} = 1/a$).

For any non-zero rational numbers a, b and natural number n , we also have: $a^{-n}/b^{-n} = (a/b)^{-n} = (b/a)^n$.

4. Worked Examples — Applying the Laws

Example 4. Simplify and write in the exponential form:

Solution. (i) $3^6 \times 3^8 \times 3^3 = 3^{6+8+3} = 3^{17}$ (using Law 1)
(ii) $5^{20} \div 5^{11} = 5^{20-11} = 5^9$ (using Law 2)

Example 5. Simplify and write in the exponential form:

Solution. (i) $(6^2)^4 = 6^{2 \times 4} = 6^8$ (using Law 3)
(ii) $(5^2)^3 \times 5^4 = 5^6 \times 5^4 = 5^{10}$ (using Laws 3, 1)
(iii) $[(-3)^3]^5 = (-3)^{15} = -(3)^{15}$ (using Law 3)

Example 6. Simplify and write in the exponential form:

Solution. (i) $(5/9)^3 \div (7/9) \dots$ using Law 5 style manipulations to reach a single exponential term, combining numerator and denominator powers step by step.

Example 7. Express each of the following rational numbers in the exponential form:

Solution. (i) $64/243 = 2^6/3^5 \dots$ expressed after prime factorisation of numerator and denominator, then simplified using Law 5 where the exponents match.
(ii) $-27/243 = -3^3/3^5 = -(1/3^2) = -1/9$ (using Laws 2, 7)

Example 8. Simplify the following:

Solution. (i) $(-4m)^3 = (-4)^3 \times m^3 = -64m^3$ (using Law 4)
(ii) $(2/3)^{-3} = 1/(2/3)^3 = (3/2)^3 = 27/8$ (using Law 7)

Example 9. Simplify the following, expressing the answer in exponential form:

Solution. As $4 = 2 \times 2 = 2^2$ and $32 = 2^5$:
(i) $2^3 \times 3^4 \times 4 = 2^3 \times 3^4 \times 2^2 = 2^5 \times 3^4$ (using Law 1)
(ii) As $125 = 5^3$ and $32 = 2^5$: $125^4 \div 3^2 = 5^{12} \div 3^2 = (5^{12}/3^2)$

Example 10. Simplify the following expressions using laws of exponents:

Solution. (i) $12^4 \times 9^3 \times 4 / (6^3 \times 8^2 \times 27)$, expressed via prime factorisation of each base into powers of 2 and 3, then combined and reduced by adding/subtracting matching exponents.
(ii) A similar expression involving 2, 3 as prime bases, simplified to a final power of 2 and/or 3.

Example 11. Simplify and write the following in the exponential form:

Solution. (i) $10^2 \times 7^0 + 3^3 \times 2^2 = 100 \times 1 + 27 \times 4 = 100 + 108 = 208$ (using $a^0 = 1$)
(ii) $4^3 \times 5^0 + 3^3 \times 2^2$ — similarly reduced using the zero-exponent law before adding the numerical results.

Example 12. By what number should we multiply $(-6)^3$ so that the product is 3^5 ?

Solution. Let the required number be x, then $(-6)^3 \times x = 3^5$ ■ $x = 3^5 / (-6)^3 = 3^5 / (-2^3 \times 3^3) = 3^2 / (-2^3) = -(3^2/2^3)$.

Example 13. Find m so that $(2/9)^3 \times (2/9)^6 = (2/9)^{2m-1}$.

Solution. $(2/9)^{3+6} = (2/9)^{2m-1}$ ■ $(2/9)^9 = (2/9)^{2m-1}$ ■ $2m - 1 = 9$ ■ $2m = 10$ ■ $m = 5$.

Exercise 4.2

1. Using laws of exponents, simplify and write the following in the exponential form:

(a) $2^7 \div 2^4$ (b) $(3/5)^3 \times (3/5)^2$ (c) $5^3 \times 5^7 \div 5^{12}$ (d) $(-7)^5 \times (-7)^{11}$

2. Simplify and write the following in the exponential form:

(a) $(2^2)^{100}$ (b) $(7^{12} \div 7^9)$ (c) $(5^3 \times 5^7) \div 5^4$

3. Simplify and write the following in the exponential form:

(a) $25^1 \div 5^3$ (b) $(-7)^0$ (c) $(3^2)^5 \div (3^7)$

4. Simplify and write the following in the exponential form:

(a) $3^3 \times 5^4$ (b) $4^8 \times 5^4$ (c) $(2^3)^{100}$

5. By what number should we multiply $3^5 \times 5^7 \div 5^1$, then find the value of x.

5. Uses of Exponents & Scientific Notation

Very large numbers are inconvenient to read, understand, and compare. To make such numbers convenient to write and read, we use powers of 10.

Scientific notation

A number is said to be in **standard (scientific) form** if it is expressed as $k \times 10^n$, where $1 \leq k < 10$ and n is a whole number, and k is called the **significand**.

Converting a number from decimal (usual) notation to standard form

- (i) Move the decimal point to the left till you get just one digit to the left of the decimal point.
- (ii) Write the given number as the product of the number so obtained and 10^n , where n is the number of places the decimal point has been moved to the left.

Converting a number from standard form to decimal (usual) notation

- (i) Write the significand and move the decimal point to the right by the number of places indicated by the exponent of 10, adding trailing zeros as necessary.

Comparing numbers in standard form

- (i) The number with the greater power of 10 is greater.
- (ii) If the powers of 10 are equal in both numbers, then compare their significands — the number with the greater significand is greater.

Worked examples

Example 1. Write the following numbers in standard (scientific) form: (i) 5.37×10^4 (ii) 7.63×10^4 (iii) 5.37×10^4

Solution. (i) $5.37 \times 10^4 = 53700$

(ii) $7.63 \times 10^4 = 76300$

(iii) $5.37 \times 10^4 = 53700$

Example 2. Which of the following numbers is the greatest? 1.45×10^{32} , -1.45×10^{32} , 1.45×10^{-32} , -1.45×10^{-32} .

Solution. Since all positive numbers are greater than all negative numbers, we can say that the greatest number is either (a) or (c). Since $32 > -32$, we can say 1.45×10^{32} is greater than 1.45×10^{-32} . So option (a) is correct.

Example 3. Express the numbers appearing in the following statements in standard form:

Solution. (i) The radius of the earth is 6,400,000 m = 6.4×10^6 m.

(ii) The speed of light is 300,000,000 m/sec = 3×10^8 m/sec.

(iii) The distance between the sun and the earth is 149,600,000,000 m = 1.496×10^{11} m.

Example 4. Compare the following numbers written in standard form: 2.7×10^{12} and 1.5×10^8 .

Solution. Since $12 > 8$, the power of 10 in 2.7×10^{12} is greater, so it is the greater number.

Exercise 4.3

1. Write the following numbers in the standard form (or scientific notation):

- (i) 5908.78 (ii) 39087.8 (iii) 3,43,000 (iv) 4,88,000

2. Write the following numbers in the expanded exponential form:

- (i) 1,205 (ii) 893,000,003 (iii) $9,083 \times 10^{11}$

3. Express the following statements in scientific notation:

- (i) The distance between the earth and the moon is 384,000,000 m.
(ii) The diameter of the sun is 1,400,000,000 m.
(iii) In a galaxy there are on an average 100,000,000,000 stars.

4. Compare the following numbers in the expanded exponential form:

$$4.3 \times 10^{14}, 3.01 \times 10^{12}, 1.4335 \times 10^{12}$$

5. Write the following numbers in the expanded form:

- (i) $4 \times 10^4 + 5 \times 10^3 + 3 \times 10^2 + 2 \times 10^1$ (ii) 3,006,194

6. Find the number from each of the following expanded forms.

6. Objective Type Questions

Solved MCQ

In the following example, choose the correct answer from the given four options.

The expression $(xyz)^3 \div (xy)^2$ is equal to:

- (a) $x^2y^2z^3$ (b) xyz^3 (c) $(xyz)^3$ (d) $x^3y^2z^2$

Solution. $(xyz)^3 \div (xy)^2 = x^3y^3z^3 \div x^2y^2 = x^{3-2}y^{3-2}z^3 = xyz^3$. **Correct answer is (b).**

Mental Maths

1. Fill in the blanks:

- (i) In the expression $(-5)^9$, exponent = ____ and base = ____.
(ii) If the base is $3/4$ and exponent is 5, then the exponential form is ____.

2. State whether the following statements are true (T) or false (F):

- (i) If a is a rational number then $a^m \times a^n = a^{m+n}$ (ii) $2^4 \times 2^{11} - 2^7 \times 2^8$ is 1

3. The value of $(-2)^{-3}$ is ____.

4. If $(100)^x = 10^4$, then the value of x is ____.

Multiple Choice Questions

3. $a \times a \times a \times b \times b \times b$ is equal to

- (a) a^3b^3 (b) $(ab)^3$ (c) 72 (d) a^4b^6

4. $(-2)^3 \times (-3)^3$ is equal to

- (a) 6^6 (b) -6^6 (c) 6^3 (d) a^4b^6

5. $(pqr)^3$ is equal to
 (a) $p^3q^3r^3$ (b) pqr^3 (c) p^3qr (d) $p^3q^3r^3$
6. $(-3/2)^{-1}$ is equal to
 (a) $3/2$ (b) $-3/2$ (c) $2/3$ (d) $-2/3$
7. $(-3/4)^{-5}$ is equal to
 (a) $-4/3$ (b) $4/3$ (c) $3/4$ (d) $4/6$
8. The value of $(5^{30} \times 5^{20}) \div (5^9)^6$ is
 (a) $-81/256$ (b) $81/256$ (c) $-243/1024$ (d) $243/1024$
9. The value of 5^{-5} in the exponential form is
 (a) 5^5 (b) 5^{-5} (c) 5^{50} (d) 5^{95}
10. The value of the expression $a^n = a^m/b^n$ does not hold when
 (a) $a = 3, b = 2$ (b) $a = -2, b = 3$ (c) $n = 0$ (d) $b = 0$
11. The value of $10^{22} + 10^{20} \div 10^{20}$ is
 (a) 2 (b) -2 (c) $1/16$ (d) 10^{42}
12. The value of $5^{-1} - 6^{-1}$ is
 (a) 10 (b) 101 (c) 30 (d) -30
13. The value of $(6^{-1} - 8^{-1})$ is
 (a) 30 (b) -30 (c) $1/24$ (d) $-1/24$
14. The value of $[(1/3)^{-2} + (1/4)^{-2}] + (1/5)^{-2}$ is
 (a) -2 (b) -1 (c) 24 (d) 1
15. If $2^3 + 1^3 = 3^x$, then the value of x is
 (a) 0 (b) 1 (c) 2 (d) 2
16. The standard form of 751.65 is
 (a) 7.5165×10^2 (b) 75.165 (c) 7.5165×10^4 (d) 7.51×10^2

Statement I-II Type Questions

In the following questions, two statements are given. Examine them carefully and choose the correct answer from the four options given below.

19. Statement I: $(1/b)^n = a^n/b^n$ is the reciprocal of a^n .

Statement II: a^n/b^n is not defined if $n = 0$.

20. Statement I: $(-3)^{-300}$ is false but Statement II is true.

Statement II: If a/b is a non-zero rational number, then $(a/b)^{-n}$ is the reciprocal of $(a/b)^n$.

21. Statement I: $324 < 165 \dots$ comparing exponential terms.

Statement II: If a, b, x, y are integers such that $x > y$, then $x^a > y^b$.

- (a) Statement I is false but Statement II is true.
 (b) Statement I is true but Statement II is false.
 (c) Both Statement I and Statement II are true.
 (d) Both Statement I and Statement II are false.

7. Chapter Summary

If a is any rational number and n is a natural number, then $a^n = a \times a \times a \times \dots$ (multiplied n times), where a is called the **base** and n is called the **exponent** (or **index**) of a^n . a^n is read as 'a raised to the power n ' or simply 'a power n .'

Law	Statement
Law 1	$a^m \times a^n = a^{m+n}$
Law 2	$a^m \div a^n = a^{m-n}$ ($m > n$)
Law 3	$(a^m)^n = a^{mn}$
Law 4	$a^n \times b^n = (ab)^n$
Law 5	$a^n/b^n = (a/b)^n$ ($b \neq 0$)
Law 6	$a^0 = 1$ ($a \neq 0$)
Law 7	$a^{-n} = 1/a^n$ ($a \neq 0$); in particular, $a^{-1} = 1/a$

Standard (Scientific) Form

A number is said to be in the standard form if it is expressed as $k \times 10^n$, where $1 \leq k < 10$ and n is a whole number, and k is called the significand.

8. Check Your Progress

- Find the value of each of the following: (i) $(-3)^3 \times 5^2$ (ii) $(-1)^{501} \times [(27)^4]$
- Simplify the following: (i) $7^3 \times 11^{14} \times 13^0/7^2 \times 11^{12}$ (ii) $(-2)^3 \times (3x)^2 \times (-xy^3)/3x^2y$
- Simplify and write the following in the exponential form: (i) $(-3)^5 \times 8^3 \times 2^5$ (ii) $9^6 \times 78^3 \times (x^2)^2$ (iii) $[(-5)^3]^4 \times 8^2 / (4^3 \times (12.5)^3)$
- If $(-3/5)^x = -27/125$, then find the value of x .
- Write the prime factorisation of the following numbers in the exponential form: (i) 24000 (ii) 14157
- Express the numbers appearing in the following statements in scientific notation: (i) The earth has 1,353,000,000 cubic km of water. (ii) The population of India was about 1,429,000,000 in 2024.
- Compare the following numbers in scientific notation: 5.976×10^{24} , 8.689×10^{23}
- If $(5/12)^8 \times (12/5)^4 = (125/1728)^x$, find x .
- If $27^{2x-1} = (243)^x$, then find the value of x .
- If $x = 1/6$, $y = 5/3$, $z = 2/4$, $w = 4/9$, then find the value of $x^3 - 3$.

Reformatted study notes based on the uploaded ICSE Mathematics VII chapter 'Exponents and Powers'. Some worked-example steps that were illegible or handwritten in the source scan have been paraphrased at the method level; verify final numeric answers against your textbook / teacher's key.